

ISEN/DAEN 427 Lecture 5

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Bayesian Inference with Beta Prior and Binomial Likelihood
Conjugate Prior, Cue Combination

BM Ch. 5, 6



“Beta”, “Gamma”

- Comes from 19th-century mathematician **Adrien-Marie Legendre**.
- He introduced:

$\Gamma(x)$ (Gamma function) and $B(x, y)$ (Beta function)

- Fun fact: chose Greek letters simply for notation:
 Γ first, then B (Beta).
- Later, the **Beta distribution** was named after the Beta function it uses.

Gamma Function Definition

The **Gamma function** $\Gamma(x)$ is defined for $x > 0$ by

$$\Gamma(x) = \int_0^{\infty} t^{x-1} e^{-t} dt.$$

It extends the factorial function to real (and complex) numbers.

- x is a parameter of the function (like an input, real or complex).
- t is the integration variable just like x or y in a regular integral.
- Think of t in the Gamma function exactly like the integration variable in the area under a curve it disappears after the integral is evaluated.

Gamma Function Key Properties

- **Factorial connection:**

$$\Gamma(n) = (n-1)! \quad \text{for } n = 1, 2, 3, \dots$$

- **Recursive property:**

$$\Gamma(x+1) = x \Gamma(x).$$

- **Special value:**

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$$

Relation Between Beta and Gamma Functions

The Beta function can be expressed in terms of Gamma functions:

$$B(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}.$$

This relationship is essential for simplifying Bayesian integrals and working with probability distributions.

- The Beta function serves as a normalizing constant in Beta and related distributions.
- The Gamma function generalizes factorials and connects deeply with probability theory.
- In Bayesian inference, Beta functions simplify posterior and marginal likelihood computations.

Beta Function Definition

The **Beta function** is defined for $\Re(x) > 0$ and $\Re(y) > 0$ as

$$B(x, y) = \int_0^1 t^{x-1} (1-t)^{y-1} dt.$$

Multiplication of Two Beta Functions

For $B(x, y)$ and $B(u, v)$:

$$B(x, y) \cdot B(u, v) = \int_0^1 \int_0^1 t^{x-1} (1-t)^{y-1} s^{u-1} (1-s)^{v-1} dt ds.$$

Using the Gamma function relation:

$$B(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)},$$

we get

$$B(x, y) B(u, v) = \frac{\Gamma(x)\Gamma(y)\Gamma(u)\Gamma(v)}{\Gamma(x+y)\Gamma(u+v)}.$$

Beta Function in Bayesian Inference

- The **Beta distribution** is defined by

$$\text{Beta}(\theta \mid \alpha, \beta) = \frac{1}{B(\alpha, \beta)} \theta^{\alpha-1} (1 - \theta)^{\beta-1}.$$

- In Bayesian inference, we often use a **Beta prior** for the probability θ in a Binomial model.
- The Beta prior is **conjugate** to the Binomial likelihood, leading to a Beta posterior.
- Posterior parameters:

$$\alpha' = \alpha + x, \quad \beta' = \beta + n - x.$$

- Beta functions appear in marginal likelihoods and normalizing constants.

Marginal Likelihood Example

Given x successes in n trials:

$$P(x) = \int_0^1 P(x \mid \theta) P(\theta) d\theta = \frac{\binom{n}{x} B(x + \alpha, n - x + \beta)}{B(\alpha, \beta)}.$$

Products and ratios of Beta functions arise naturally when integrating out θ .

Example

Scenario

A car-sharing company lets drivers drop off and pick up cars anywhere in the city. You want to find the probability of having at least one car near your house at any time.

- Over 3 days, you record the number of nearby cars: $\mathbf{x} = [3, 4, 1]$
- Assume the number of cars follows a Poisson distribution (rate λ)

Step 1: Estimate λ (Maximum Likelihood)

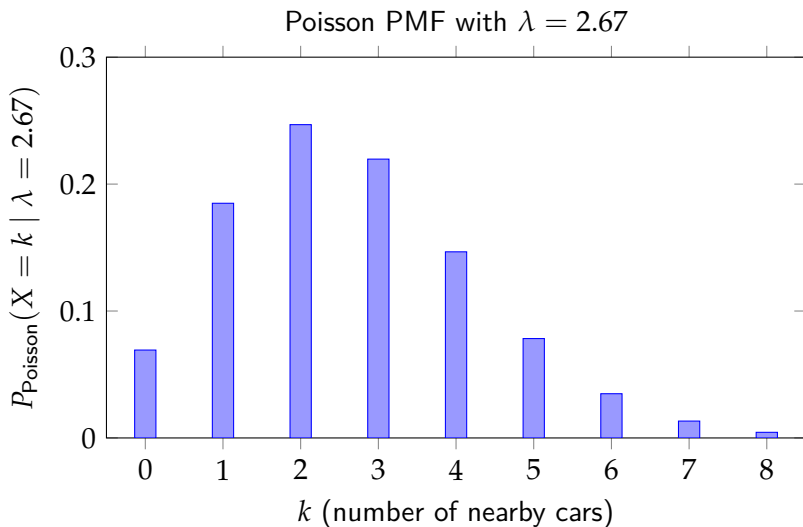
$$\lambda = \frac{3+4+1}{3} \approx 2.67$$

Step 2: Probability of at least one car

$$P(x > 0 \mid \lambda = 2.67) = 1 - P(x = 0 \mid \lambda = 2.67) = 1 - e^{-2.67} \approx 0.93$$

About 93% chance that a car is available near your house.

Poisson MLE Prediction (PMF)



$$P(X > 0) = 1 - P(X = 0) = 1 - e^{-2.67} \approx 0.93$$

With only 3 data points, λ might not be accurate. Bayes allows us to *combine prior knowledge* with the data to get a more cautious estimate.

Step 1: Add a Prior (Gamma Distribution Γ)

$$\alpha = 2, \quad \beta = 2$$

Step 2: Update with Data $\rightarrow$ Posterior Parameters

$$\alpha' = \alpha + \sum x_i = 2 + (3 + 4 + 1) = 10$$

$$\beta' = \beta + n = 2 + 3 = 5$$

Step 3: Posterior Probability

Using the negative binomial distribution:

$$P(x > 0 \mid \mathbf{x}) = 1 - P(x = 0 \mid \mathbf{x}) = 1 - NB(0 \mid 10, \frac{5}{1+5}) \approx 0.84$$

Bayesian estimate is 84% (more conservative than 93%).

α, β of the Beta Distribution

- The Beta distribution has two shape parameters:
 - α (alpha) First shape parameter
 - Interpreted as successes + 1 in Bayesian contexts
 - Larger $\alpha \Rightarrow$ distribution skews toward 1
 - β (beta) Second shape parameter
 - Interpreted as failures + 1
 - Larger $\beta \Rightarrow$ distribution skews toward 0

- **PDF:**

$$f(x; \alpha, \beta) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}$$

where

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}$$

- **Mean:** $\frac{\alpha}{\alpha + \beta}$ **Variance:** $\frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$

- **Poisson MLE:** Good point estimate with lots of data.
 - **Bayesian Posterior:** Accounts for uncertainty with little data.
 - Real-world problems often need both data & prior knowledge.
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- Beta Distribution Applet/Calculator
 - Beta Distribution Mathlets

Bayesian Inference with Binomial Likelihood

- Assume observations $x_1, \dots, x_N$ are binary $x_i \in \{0, 1\}$ and $s \in [0, 1]$.
- Given a likelihood $\mathbb{P}(x_i|s)$, conditional independence implies:

$$\begin{aligned}\mathbb{P}(x_1, \dots, x_N | s) &= \prod_{i=1}^N \mathbb{P}(x_i | s) \\ &= \prod_{i=1}^N s^{x_i} (1 - s)^{1-x_i} \\ &= s^{\sum_{i=1}^N x_i} (1 - s)^{N - \sum_{i=1}^N x_i}\end{aligned}$$

where $\sum_{i=1}^N x_i$ is the total number of ones ("successes") and $N - \sum_{i=1}^N x_i$ are the total number of zeros ("failures").

Bayesian Inference with Binomial Likelihood

- The likelihood can alternatively be written in terms of the Binomial distribution for $y = \sum_{i=1}^N x_i$:

$$\mathbb{P}(y|s) = \binom{N}{y} s^y (1-s)^{N-y}$$

where $y \in \{0, 1, \dots, N\}$.

- Recall that

$$\mathbb{E}[y] = Ns \quad \text{Var}[y] = Ns(1-s)$$

Beta vs Binomial in Bayesian Inference

- **Binomial distribution** models **data**:
 - Number of successes in n trials given probability p .
 - Example: How many heads in 10 coin flips?
- **Beta distribution** models **beliefs about** p :
 - Represents uncertainty over the unknown probability itself.
 - Example: How likely is $p = 0.7$?

In Bayesian inference, we often want to infer p from data.
The Beta distribution is ideal for this.

Conjugacy: Beta is conjugate prior of the Binomial.

$$p \sim \text{Beta}(\alpha, \beta), \quad \text{data} \sim \text{Binomial}(n, p)$$

$$p \mid \text{data} \sim \text{Beta}(\alpha + \text{successes}, \beta + \text{failures})$$

- Posterior is also Beta $\Rightarrow$ simple closed-form updates.
- Flexible: α and β can express uniform, peaked, or skewed beliefs.
- Binomial is used as **likelihood**, Beta is used as **prior and posterior**.

Use Binomial for data, Beta for beliefs.
Easily update with new evidence.

Note: *What is Conjugacy?*

- In Bayesian inference:

$$\text{Posterior} \propto \text{Likelihood} \times \text{Prior}$$

- A prior and likelihood are **conjugate** if the posterior has the **same distribution family** as the prior.

Example: Beta–Binomial Conjugacy

$$p \sim \text{Beta}(\alpha, \beta), \quad \text{Data} \sim \text{Binomial}(n, p)$$

$$p \mid \text{data} \sim \text{Beta}(\alpha + \text{successes}, \beta + \text{failures})$$

- Posterior stays in the Beta family $\rightarrow$ simple math.
- Easy to update beliefs as new data arrives.
- No complicated integrals needed.

Prior: Beta distribution

- The beta distribution with parameters $\alpha > 0$ and $\beta > 0$ is of the form:

$$\mathbb{P}(s) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} s^{\alpha-1} (1 - s)^{\beta-1}$$

- The mean and variance: $\mathbb{E}[s] = \frac{\alpha}{\alpha + \beta}$ and $\text{Var}[s] = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$

The Beta distribution is a type of probability distribution.

- It describes probabilities for values between 0 and 1 (like percentages).
- It is controlled by two numbers: α and β .

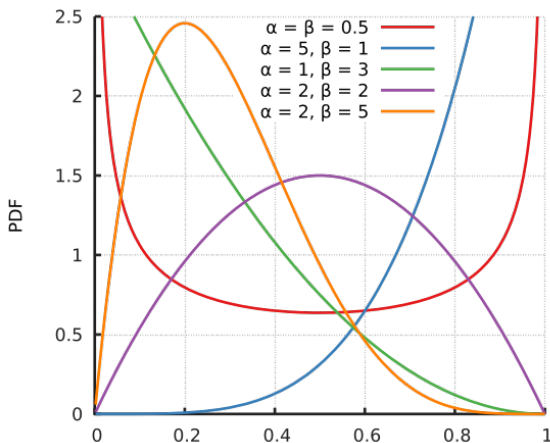
The Beta distribution is great for modeling situations like:

- The probability of success in experiments (e.g., shooting hoops).
- Estimating proportions when you have some data but not too much.
- Bayesian statistics (used to update beliefs as new data comes in).

Different Shapes

Depending on α and β , the shape changes:

- $\alpha = \beta = 1 \rightarrow$ Flat (every value equally likely).
- $\alpha > \beta \rightarrow$ Skewed toward 1 (more likely high values).
- $\alpha < \beta \rightarrow$ Skewed toward 0 (more likely low values).



Learning as Inference

Assume the a priori distribution is **beta** with parameters $\alpha > 0$ and $\beta > 0$ and $y = \sum_{i=1}^N x_i$ then

$$\begin{aligned}\mathbb{P}(s|y) &= \kappa \mathbb{P}(y|s) \mathbb{P}(s) \\&= \kappa \binom{N}{y} s^y (1-s)^{N-y} \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha) \Gamma(\beta)} s^{\alpha-1} (1-s)^{\beta-1} \\&= \kappa \binom{N}{y} \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha) \Gamma(\beta)} s^{y+\alpha-1} (1-s)^{N+\beta-y-1}\end{aligned}$$

where

$$\kappa = \frac{1}{\int_0^1 \mathbb{P}(y|s) \mathbb{P}(s) ds} = \frac{1}{\mathbb{P}(y)}$$

Computing the Posterior

- After recording y ones ("successes") out of $N > 1$ the posterior distribution is **beta** with parameters α' and β' , i.e.:

$$\mathbb{P}(s|y) = \frac{\Gamma(\alpha' + \beta')}{\Gamma(\alpha')\Gamma(\beta')} s^{\alpha'-1} (1-s)^{\beta'-1}$$

- Thus,

$$\mu_{post} = \frac{\alpha'}{\alpha' + \beta'} = \frac{y + \alpha}{N + \alpha + \beta}$$

$$\begin{aligned}\sigma_{post} &= \frac{\alpha' \beta'}{(\alpha' + \beta')^2 (\alpha' + \beta' + 1)} \\ &= \frac{(y + \alpha)(N - y + \beta)}{(N + \alpha + \beta)^2 (N + \alpha + \beta + 1)}\end{aligned}$$

Computing the Posterior

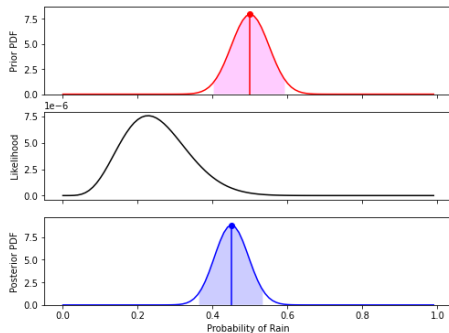


Figure 1: Posterior with Beta Prior and Binomial Likelihood
 $s = \mathbb{P}(\text{"rain"})$

Connection to Reinforcement Learning

- With $\alpha = \beta = 1$, an additional $N + 1$ observation leads to the posterior mean

$$\mu_{post,N+1} = \frac{y + x_{N+1} + 1}{N + 1 + 2}$$

- A recursive update equation for the posterior mean

$$\begin{aligned}\mu_{post,N+1} &= \frac{y + 1}{N + 2} \frac{N + 2}{N + 3} + x_{N+1} \frac{1}{N + 3} \\ &= \mu_{post,N} \left(1 - \frac{1}{N + 3}\right) + x_{N+1} \frac{1}{N + 3} \\ &= \mu_{post,N} + \frac{1}{N + 3} (x_{N+1} - \mu_{post,N})\end{aligned}$$

Connection to Reinforcement Learning

- This equation resembles the *Rescorla-Wagner rule* in reinforcement learning.
- New information or “surprise” is $x_{N+1} - \mu_{post,N}$.
- The factor $\frac{1}{N+3}$ acts as the learning rate.

Learning happens through the difference between
expected and **actual** outcomes.

- If the outcome is **better than expected**, the prediction increases.
- If the outcome is **worse than expected**, the prediction decreases.
- Over time, the agent learns to make more accurate predictions.

Rescorla–Wagner Rule: Formula

$$V_{t+1} = V_t + \alpha(\lambda - V_t)$$

- V_t = current prediction (expected reward)
- λ = actual outcome
- α = learning rate ($0 < \alpha \leq 1$)
- $\lambda - V_t$ = **prediction error**

Example: If a student expects 60% but scores 80%, the prediction error is +20%, leading to an upward update.

Cue Combination

- Inference based on two *different* types of measurements (cues).
- How should cues be processed in order to optimize inference?
- Experimental evidence indicates the human brain **combines** cues.

Figure 2: Testing Visual and Auditory Cue Combination

- Let x_1 and x_2 denote the observation values (cues). We assume

$$\mathbb{P}(x_1, x_2 | s) = \mathbb{P}(x_1 | s) \mathbb{P}(x_2 | s)$$

- Let us derive the posterior when

$$\begin{aligned} & \mathbb{P}(x_i | s) \\ &= \\ & \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left(-\frac{1}{2\sigma_i^2}(x_i - s)^2\right) \end{aligned}$$

Figure 3: Generative Model

for $i = 1, 2$.

Posterior Distribution

- Similarly to derivation in Lecture 4, it can be shown that:

$$\mathbb{P}(s|x_1, x_2) = \frac{1}{\sqrt{2\pi}\sigma_{post}} \exp\left(-\frac{1}{2\sigma_i^2}(s - \mu_{post})^2\right)$$

with

$$\mu_{post} = \frac{\frac{1}{\sigma_s^2}}{\frac{1}{\sigma_s^2} + \frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2}} \mu_s + \frac{\frac{1}{\sigma_1^2}}{\frac{1}{\sigma_s^2} + \frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2}} x_1 + \frac{\frac{1}{\sigma_2^2}}{\frac{1}{\sigma_s^2} + \frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2}} x_2$$

$$\sigma_{post} = \frac{1}{\frac{1}{\sigma_s^2} + \frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2}}$$

Posterior Distribution

Figure 4: Posterior Distribution from Cue Combination